Time Series Data Analysis and Forecasting

MSDS

Module 2

Topic Covered

- Introduction to time series regression model
- Least Squares Estimation
- statistical inference
- Prediction
- Variable selection
- regression models
- Classical decomposition, X11, SEATS, STL
- Forecasting with decomposition

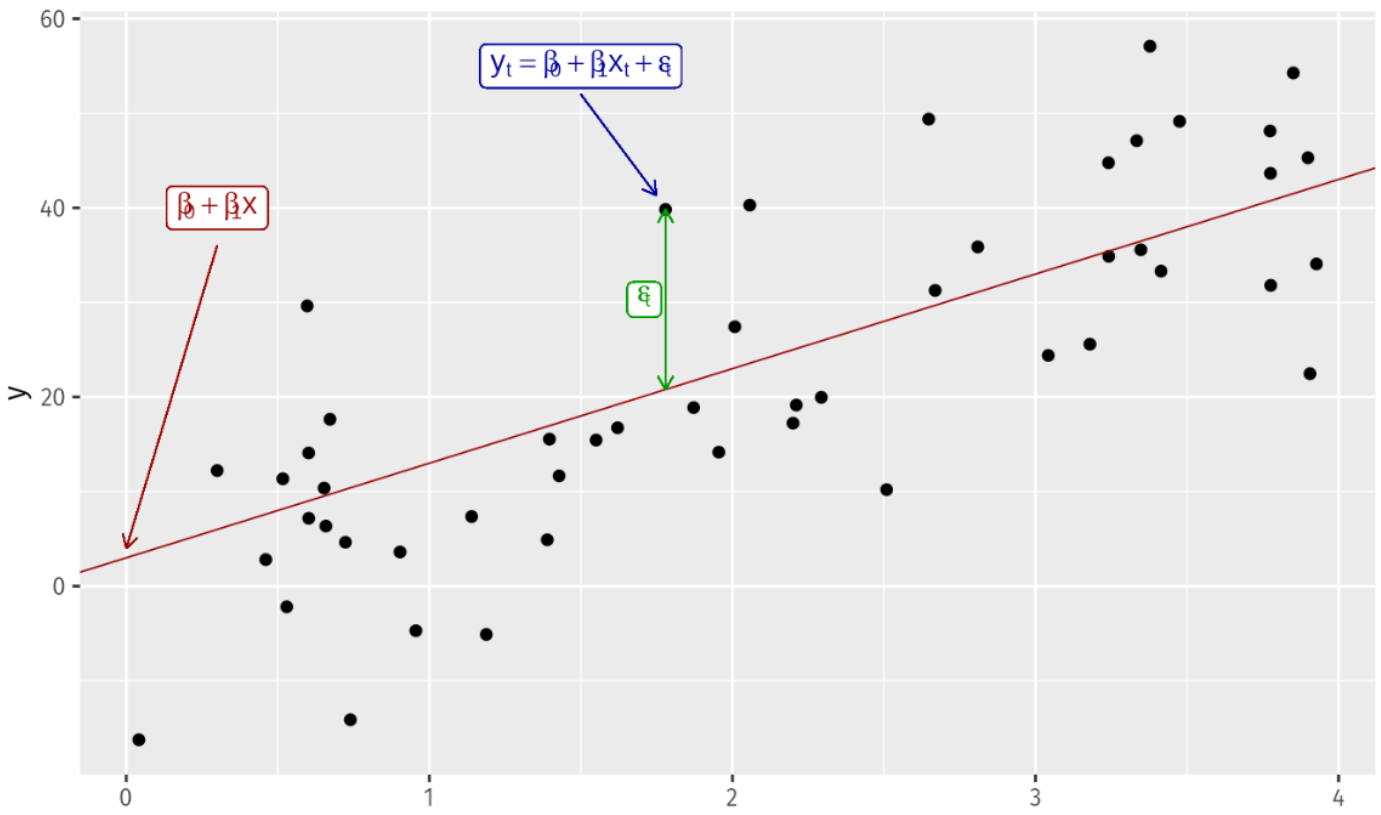
Introduction to Regression Model

- For regression analysis to be performed, data has to be stationary.
- Equation has to be rewritten in such a form that indicates a relationship among stationary variables.
- If a series is stationary then we can model it via an equation with fixed coefficents estimated from the past data
- concept is that forecast the time series of interest y assuming that it has a linear relationship with other time series x .
- The **forecast variable** y is sometimes also called the regressand, dependent or explained variable. The **predictor variables** x are sometimes also called the regressors, independent or explanatory variables.

Simple Linear Regression

- Regression model allows for a linear relationship between the forecast variable y and a single predictor variable x,

$$y_t = \beta_0 + \beta_1 x_t + \varepsilon_t.$$



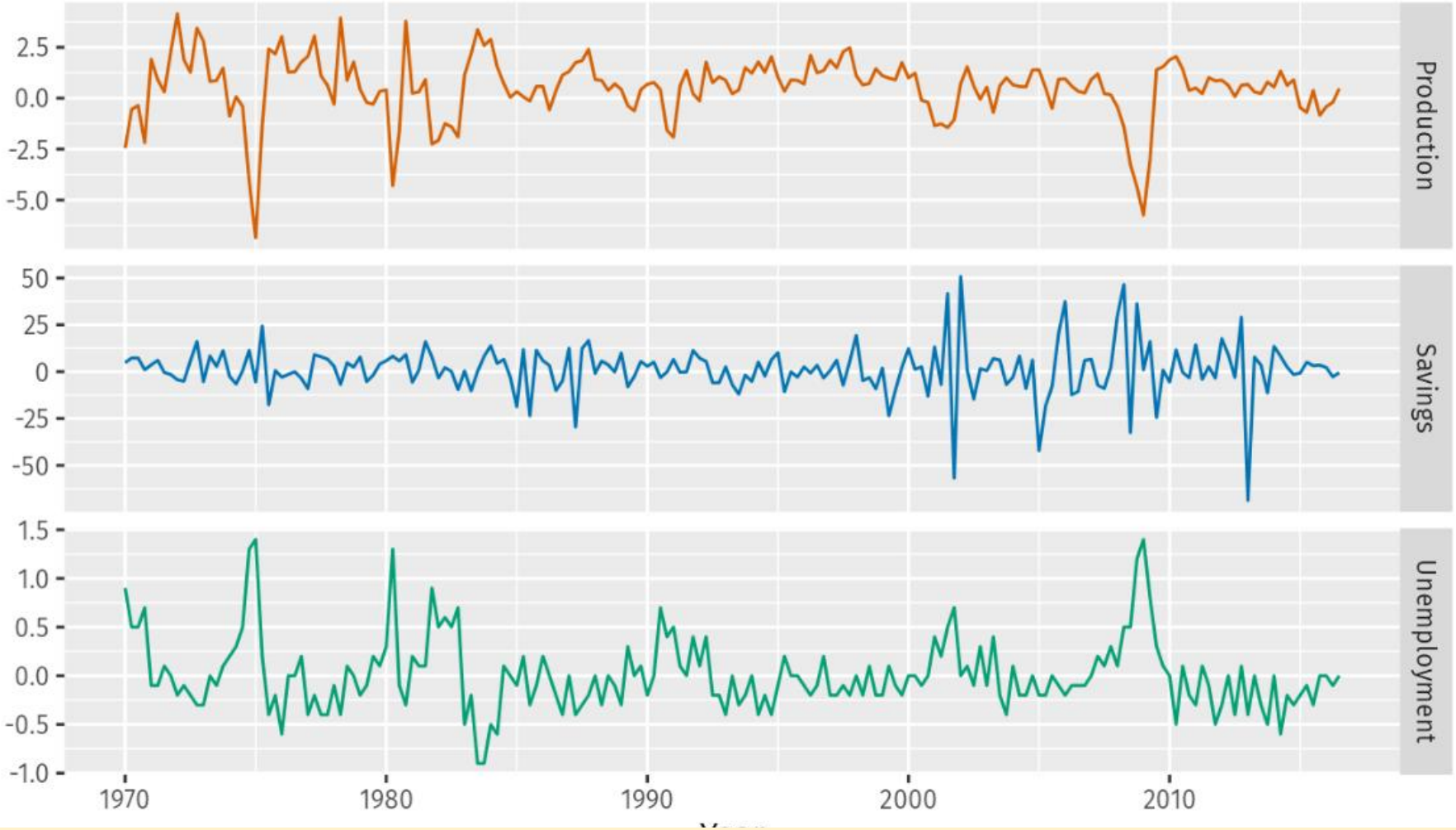
Multiple Linear Regression

- When there are two or more predictor variables, the model is called a **multiple regression model**. The general form of a multiple regression model is

$$y_t = \beta_0 + \beta_1 x_{1,t} + \beta_2 x_{2,t} + \cdots + \beta_k x_{k,t} + \varepsilon_t,$$

  - Where y is predictor variable to be forecast and $x_1, x_2 \ldots.,\ x_k$ are the k predictor variable.
  - Each of predictor variable must be numerical.
  - $x_1, x_2 \ldots.,\ x_k$ coefficient measure the effect of each predictor after taking intro account the effects of all other predictors in model.
  - Coefficient measure the marginal effects of predictor variable.

Example: US consumption expenditure



Quarterly percentage changes in industrial production and personal savings and quarterly changes in the unemployment rate for the US over the period 1970Q1-2016Q3

Least Square estimation(1)

- We have a collection of observation but the values of the coefficient β_0, β_1,… β_k are not known.
- least squares principle provides a way of choosing the coefficients effectively by minimising the sum of the squared errors.

$$\sum_{t=1}^{T} \varepsilon_t^2 = \sum_{t=1}^{T} (y_t - \beta_0 - \beta_1 x_{1,t} - \beta_2 x_{2,t} - \cdots - \beta_k x_{k,t})^2.$$

- This is called **least squares** estimation because it gives the least value for the sum of squared errors.
- Finding the best estimates of the coefficients is often called “fitting” the model to the data, or sometimes “learning” or “training” the model.
- The estimated coefficients are given by notation $\widehat{\beta_0}, \widehat{\beta_1}, , \ldots \widehat{\beta_k},$

Example: US consumption expenditure

A multiple linear regression model for US consumption is

$$y_t = \beta_0 + \beta_1 x_{1,t} + \beta_2 x_{2,t} + \beta_3 x_{3,t} + \beta_4 x_{4,t} + \varepsilon_t,$$

where y is the percentage change in real personal consumption expenditure, x_1 is the percentage change in real personal disposable income, x_2 is the percentage change in industrial production, x_3 is the percentage change in personal savings and x_4 is the change in the unemployment rate.

R code for regression modeling

```
fit.consMR <- tslm(
  Consumption ~ Income + Production + Unemployment + Savings,
  data=uschange)
summary(fit.consMR)
```

Fitted Values

- Predictions of y can be obtained by using the **estimated coefficients** in the regression equation and setting the error term to zero.
- Note that these are predictions of the data used to estimate the model, not genuine forecasts of future values of y

$$\hat{y}_t = \hat{\beta}_0 + \hat{\beta}_1 x_{1,t} + \hat{\beta}_2 x_{2,t} + \cdots + \hat{\beta}_k x_{k,t}.$$

Goodness of Fit

- A common way to summarise how well a linear regression model fits the data is via the **coefficient of determination**, or R^2
- This can be calculated as the square of the correlation between the observed values y and the predicted $\hat{y}$ values.
- it reflects the proportion of variation in the forecast variable that is accounted for (or explained) by the regression model.

$$R^2 = \frac{\sum(\hat{y}_t - \bar{y})^2}{\sum(y_t - \bar{y})^2},$$

Standard Error of Regression

- Another measure of how well the model has fitted the data is the standard deviation of the residuals, which is often known as the "residual standard error"
- Calculated by equation,

$$\hat{\sigma}_e = \sqrt{\frac{1}{T-k-1}\sum_{t=1}^{T} e_t^2},$$

- K is number of predictor

Evaluating the Regression model

The differences between the observed y values and the corresponding fitted $\hat{y}$ values are the training-set errors or "residuals" defined as,

$$\begin{aligned} e_t &= y_t - \hat{y}_t \\ &= y_t - \hat{\beta}_0 - \hat{\beta}_1 x_{1,t} - \hat{\beta}_2 x_{2,t} - \cdots - \hat{\beta}_k x_{k,t} \end{aligned}$$

for $t = 1, \ldots, T$. Each residual is the unpredictable component of the associated observation.

The residuals have some useful properties including the following two:

$$\sum_{t=1}^{T} e_t = 0 \quad \text{and} \quad \sum_{t=1}^{T} x_{k,t} e_t = 0 \qquad \text{for all } k.$$

Forecasting with Regression(1)

Recall that predictions of y can be obtained using

$$\hat{y}_t = \hat{\beta}_0 + \hat{\beta}_1 x_{1,t} + \hat{\beta}_2 x_{2,t} + \cdots + \hat{\beta}_k x_{k,t},$$

which comprises the estimated coefficients and ignores the error in the regression equation. Plugging in the values of the predictor variables $x_{1,t}, \ldots, x_{k,t}$ for $t = 1, \ldots, T$ returned the fitted (training-sample) values of y. What we are interested in here, however, is forecasting *future* values of y.

Forecasting with Regression(2)

There are two types-

- **<u>Ex-ante Forecast</u>**

-Those forecast that are made using only the information that is available in advance.

-For example, the percentage change in US consumption for quarters following the end of the sample, should only use information that was available *up to and including* 2016 Q3.

-These are made in advance using whatever information is available at the time. Model requires forecasts of the predictors.

- **<u>ex-post forecasts</u>**

-Those that are made using later information on the predictors.

-For example, ex-post forecasts of consumption may use the actual observations of the predictors, once these have been observed.

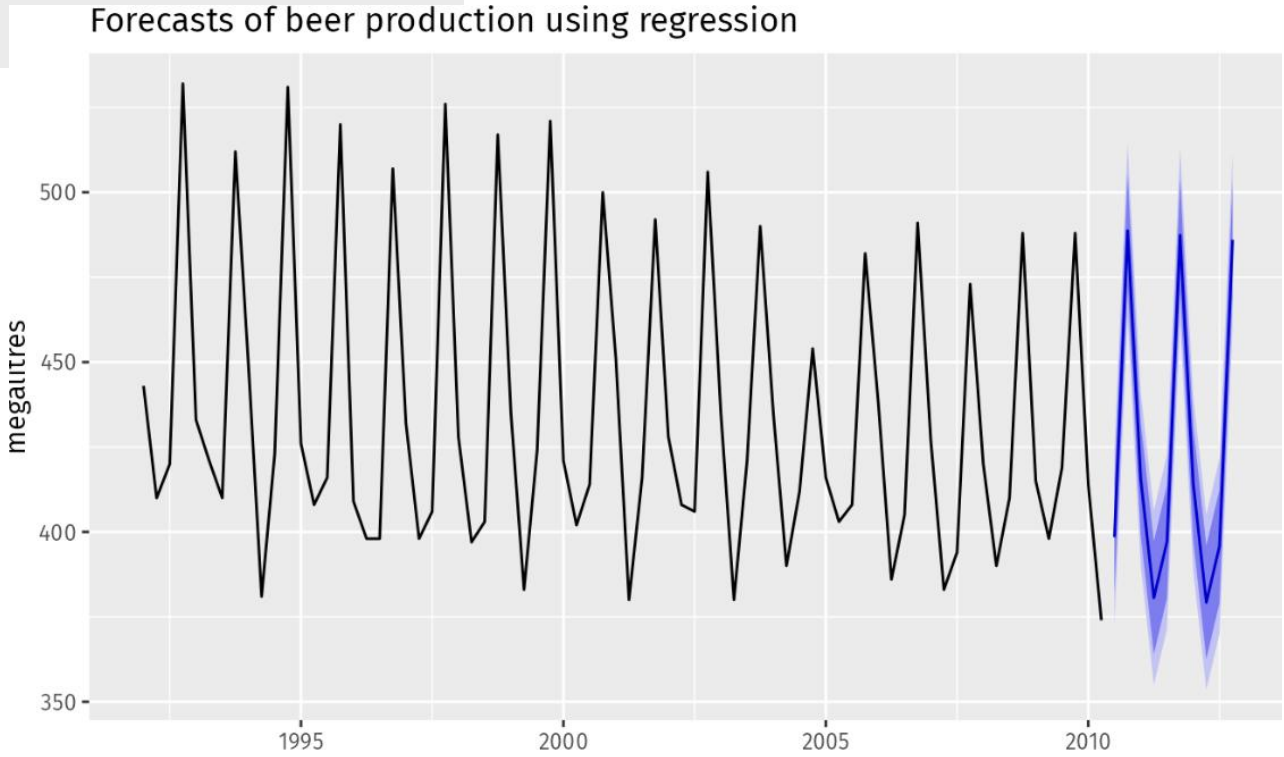
-These are not genuine forecasts, but are useful for studying the behaviour of forecasting models.

Building a predictive regression model

- The great advantage of regression models is that they can be used to capture important relationships between the forecast variable of interest and predictor variables.
- A major challenge is that in order to generate **ex-ante forecasts**, the model requires future values of each predictor.
- If scenario based forecasting is of interest then these models are extremely useful.

Forecasting Using Regression model: Australian beer production data

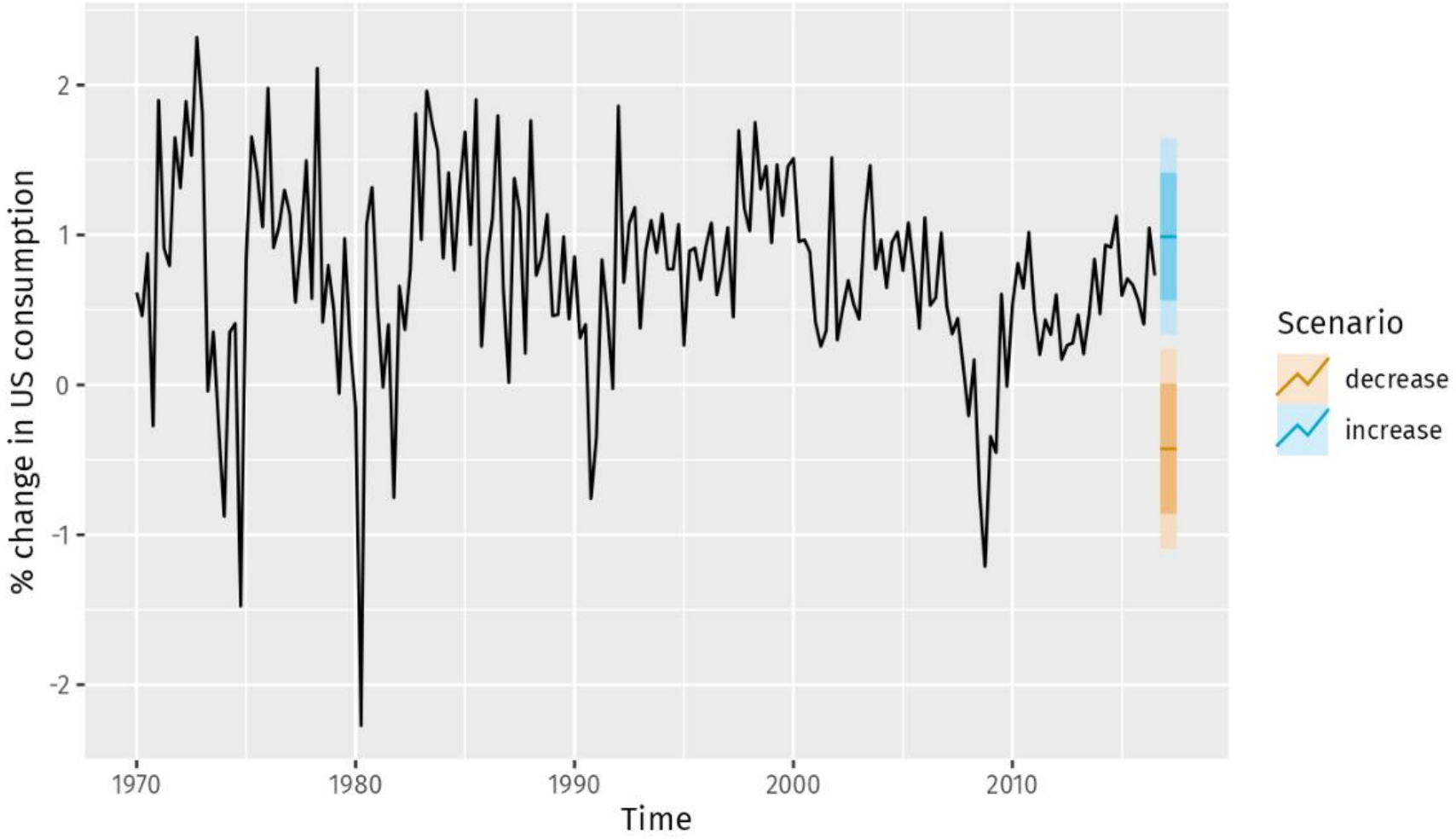
```
beer2 <- window(ausbeer, start=1992)
fit.beer <- tslm(beer2 ~ trend + season)
fcast <- forecast(fit.beer)
autoplot(fcast) +
  ggtitle("Forecasts of beer production using regression") +
  xlab("Year") + ylab("megalitres")
```



Scenario Based forecasting

- forecaster assumes possible scenarios for the predictor variables that are of interest.
- **For example**- a US policy maker may be interested in comparing the predicted change in consumption when there is a constant growth of 1%
- 0.5% respectively for income and savings with no change in the employment rate, versus a respective decline of 1% and 0.5%, for each of the four quarters following the end of the sample.
- prediction intervals for scenario based forecasts do not include the uncertainty associated with the future values of the predictor variables.
- They assume that the values of the predictors are known in advance.

Scenario Based forecasting: uschange data



Time Series Decomposition

Time Series Decomposition

- Time series data can exhibit a variety of patterns, and it is often helpful to **split a time series** into several components, each representing
- While decomposing a time series into components, combine the trend and cycle into a single **trend-cycle** component (sometimes called the **trend** for simplicity).
- A time series as comprising three components: a trend-cycle component, a seasonal component, and a remainder component

Time Series Components

If we assume an additive decomposition, then we can write

$$y_t = S_t + T_t + R_t,$$

where y_t is the data, S_t is the seasonal component, T_t is the trend-cycle component, and R_t is the remainder component, all at period t. Alternatively, a multiplicative decomposition would be written as

$$y_t = S_t \times T_t \times R_t.$$

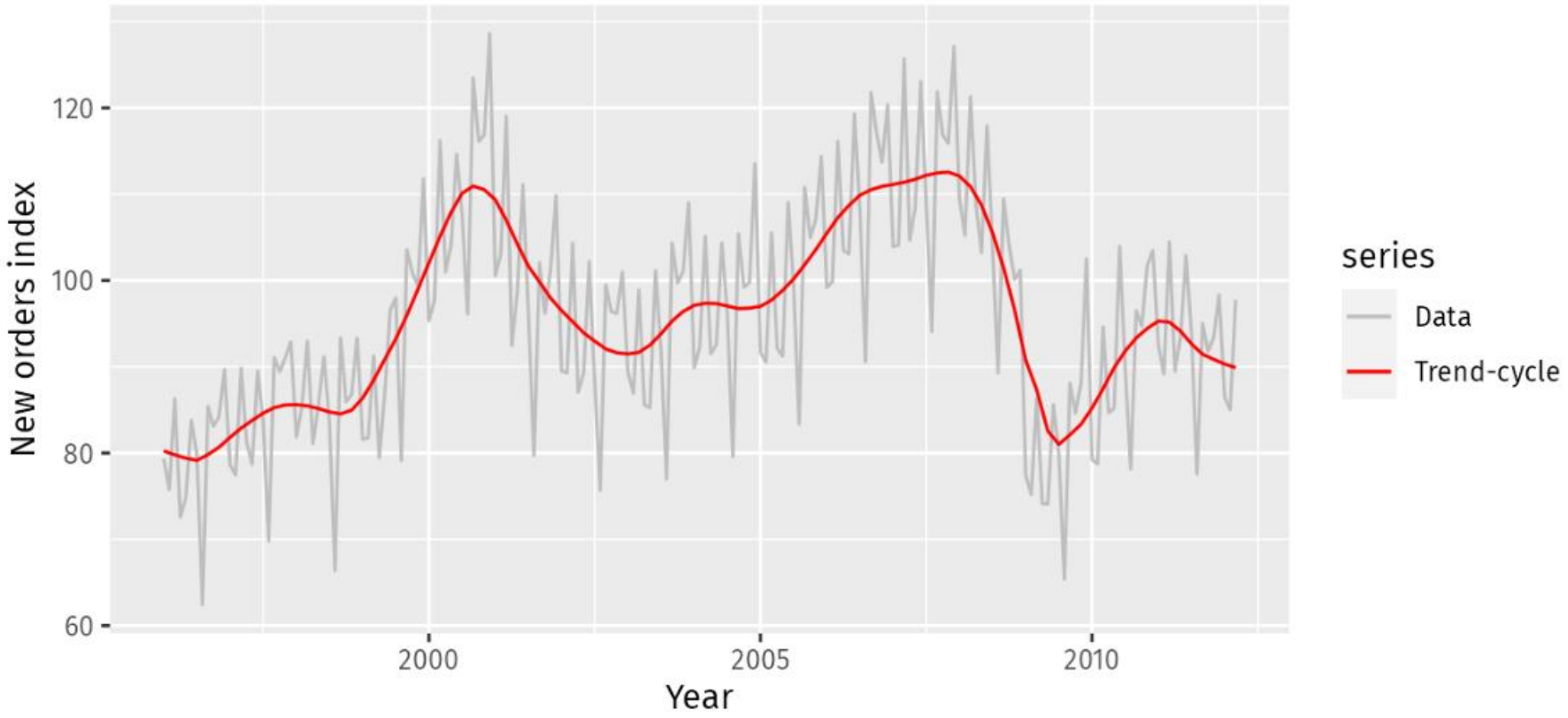
Time Series Components

- **Additive decomposition** is most appropriate if the magnitude of the seasonal fluctuations, or the variation around the trend-cycle, does not vary with the level of the time series.
- When variation in the seasonal pattern, or variation around the trend-cycle, appears to be proportional to the level of the time series, then a **multiplicative decomposition** is more appropriate.
- Multiplicative decompositions are common with economic time series.
- When a log transformation has been used, this is equivalent to using a multiplicative decomposition because

$$y_t = S_t \times T_t \times R_t \quad \text{is equivalent to} \quad \log y_t = \log S_t + \log T_t + \log R_t.$$

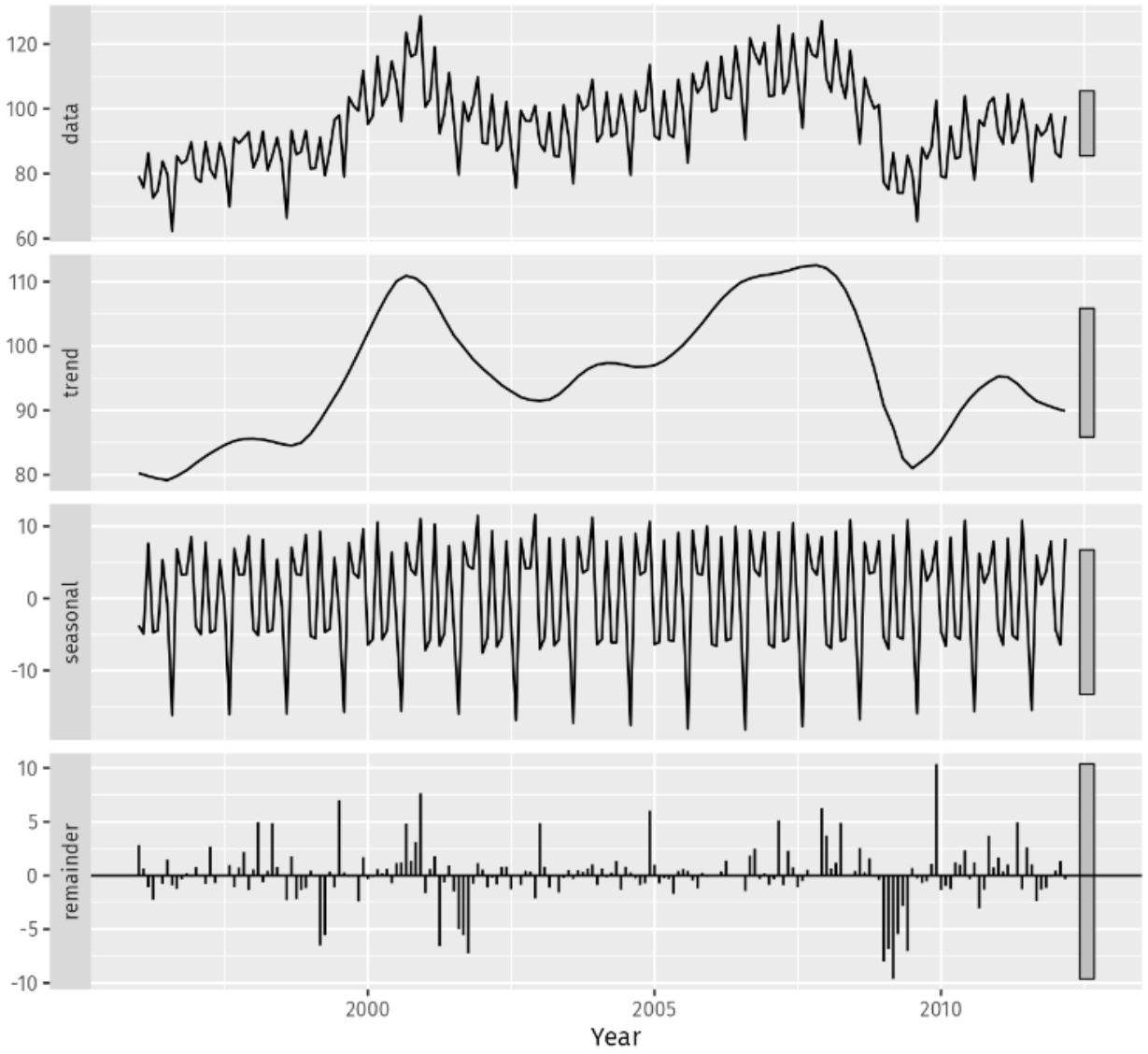
Example: Electrical equipment manufacturing

- Data show number of new orders for electrical equipment (computer, electronic and optical products) in the Euro area (16 countries).
- data have been adjusted by working days and normalised so that a value of 100 corresponds to 2005



Example: Electrical equipment manufacturing

- Figure shows **additive** decomposition
- Method used to decompose is STL
- three components are shown separately in **bottom three** panels of Figure
- These components can be added together to **reconstruct the data** in top
- **seasonal component** changes slowly over time, so that any two consecutive years have similar patterns, but years far apart may have different seasonal patterns.

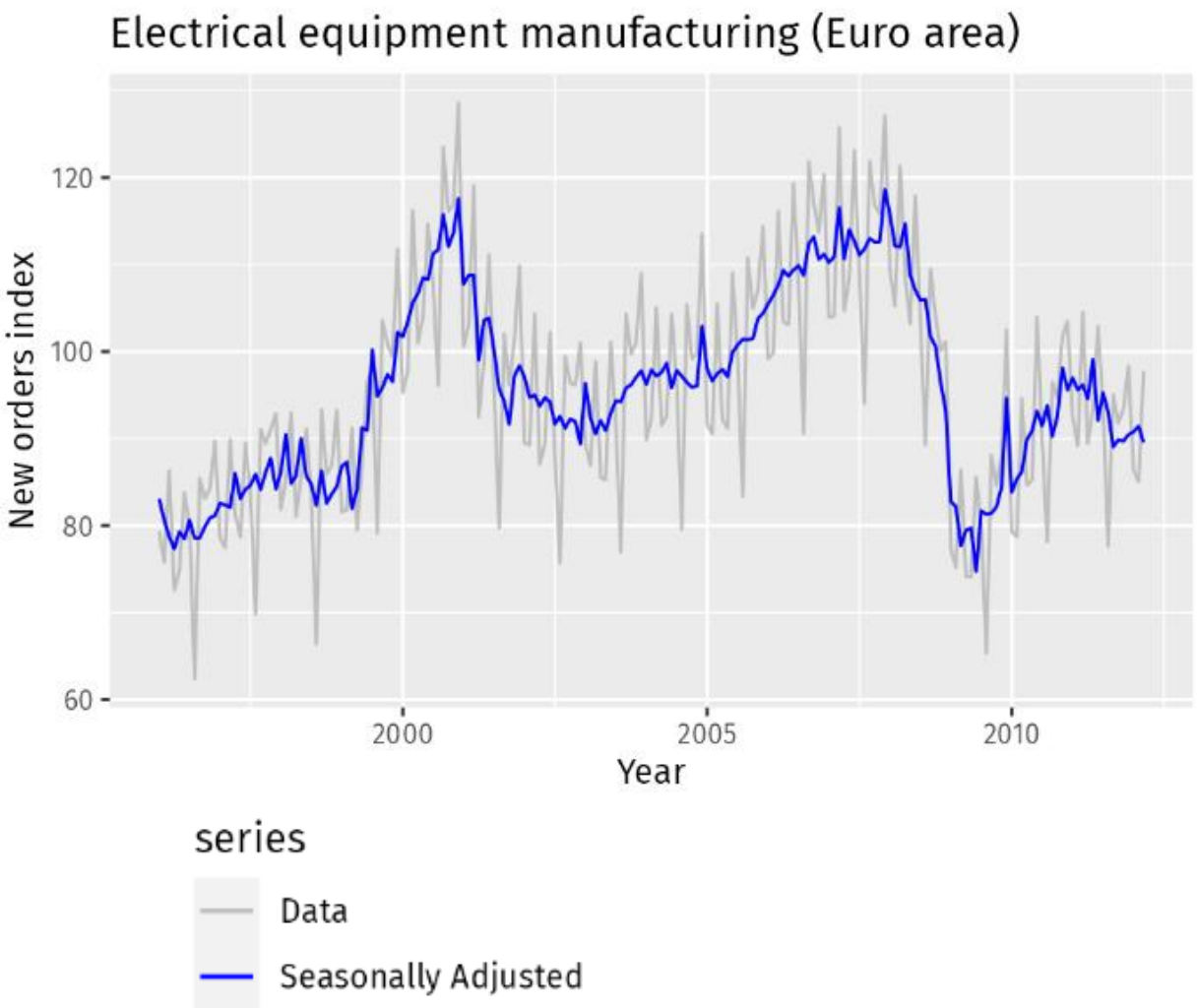


Seasonally adjusted data

- If the seasonal component is removed from the original data, the resulting values are the "seasonally adjusted" data.
- For an additive decomposition, the seasonally adjusted data are given by—

  $y_t - S_t$

- for multiplicative data, the seasonally adjusted values are obtained using y_t / S_t
- If variation due to seasonality is not of primary interest, seasonally adjusted series can be useful.



X11 Decomposition

- X11 method originated in the US Census Bureau and Statistics Canada.
- Based on classical decomposition, but includes many extra features that overcome drawbacks of classical decomposition.
- Trend-cycle estimates are available for all observations including end points, and seasonal component is allowed to vary slowly over time.
- X11 also has some sophisticated methods for handling trading day variation, holiday effects and the effects of known predictors.
- It handles additive and multiplicative decomposition.
- process is automatic and tends to be robust to outliers and level shifts in time series.

Usage-

- It is useful in seasonal plots and seasonal sub-series plots of the seasonal component.
- help us to visualise variation in the seasonal component over time

SEATS Decomposition

- “SEATS” stands for “Seasonal Extraction in ARIMA Time Series
- This procedure was developed at the Bank of Spain, and is now widely used by government agencies around the world.
- The procedure works only with quarterly and monthly data.
- So seasonality of other kinds, such as daily data, or hourly data, or weekly data, require an alternative approach.

STL Decomposition

- It is versatile and robust method for decomposing time series.
- STL is an acronym for “Seasonal and Trend decomposition using Loess”
- while Loess is a method for estimating nonlinear relationships.
- developed by R. B. Cleveland, Cleveland, McRae, & Terpenning

Advantage-

•Unlike SEATS and X11, STL will handle any type of seasonality, not only monthly and quarterly data.

•seasonal component is allowed to change over time, and the rate of change can be controlled by the user.

•smoothness of trend-cycle can also be controlled by the user.

•It can be robust to outliers (i.e., the user can specify a robust decomposition), so that occasional unusual observations will not affect the estimates of the trend-cycle and seasonal components. They will, however, affect the remainder component.

Disadvantage of STL Decomposition

- it does not handle trading day or calendar variation automatically, and it only provides facilities for additive decompositions.
- It is possible to obtain a multiplicative decomposition by first taking logs of the data, then back-transforming the components.
- Decompositions between additive and multiplicative can be obtained using a Box-Cox transformation of data with $0<\lambda<1$.
- A value of **$\lambda=0$** corresponds to the **multiplicative decomposition** while **$\lambda=1$** is equivalent to an **additive decomposition**.

STL Decomposition

- Two main parameters to be chosen when using STL are trend cycle window (**t.window**) and the seasonal window (**s.window**).
- These control how rapidally the trend cycle and seasonal components can change.
- Smaller values allow for more rapid changes
- Both t.window and s.window should be **odd numbers**.
- t.window is number of **consecutive observations** to be used when estimating the trend cycle
- s.window is number of **consecutive years** to be used inestimating each value in seasonal; component.

Measuring Strength of Trend

- A time series decomposition can be used to measure strength of trend and seasonality in a time series
- Decomposition is written as, $\boldsymbol{y_t = T_t + S_t + R_t}$
- For strongly trended data, the seasonality adjusted data should have much more variation than the remainder component.
- The **strength of trend** is defined as

$$\boldsymbol{F_t = \max\left(0, 1 - \frac{Var(R_t)}{Var(T_t + R_t)}\right)}$$

- This gives a measure of strength of the trend between 0 and 1.
- Because variance of the remainer occasionally be even larger than the variance of seasonality adjusted data.
- Minimum possible value of strength($\boldsymbol{F_t}$) is zero.

Measuring Strength of seasonality

- The strength of seasonality is defined with respect to **detrended data** rather than the seasonally adjusted data:
- The **strength of seasonality** is defined as

$$\boldsymbol{F_s = \max(0, 1 - \frac{Var(R_t)}{Var(S_t + R_t)})}$$

- A series with seasonal strength $\boldsymbol{F_s}$ close to 0 exhibits almost no seasonality
- while a series with strong seasonality will have $\boldsymbol{F_s}$ close to 1 because $\boldsymbol{Var(R_t)}$ will be much smaller than $\boldsymbol{Var(S_t + R_t)}$

Forecasting with decomposition

- While decomposition is primarily useful for studying time series data, and exploring historical changes over time, it can also be used in **forecasting**.
- Decomposed series is

$$\boldsymbol{y_t = \widehat{S_t} + \widehat{A_t}}$$

- Where $\widehat{A_t} = \widehat{T_t} + \widehat{R_t}$, is the **seasonally adjusted component**. If a multiplicative decomposition is used , write

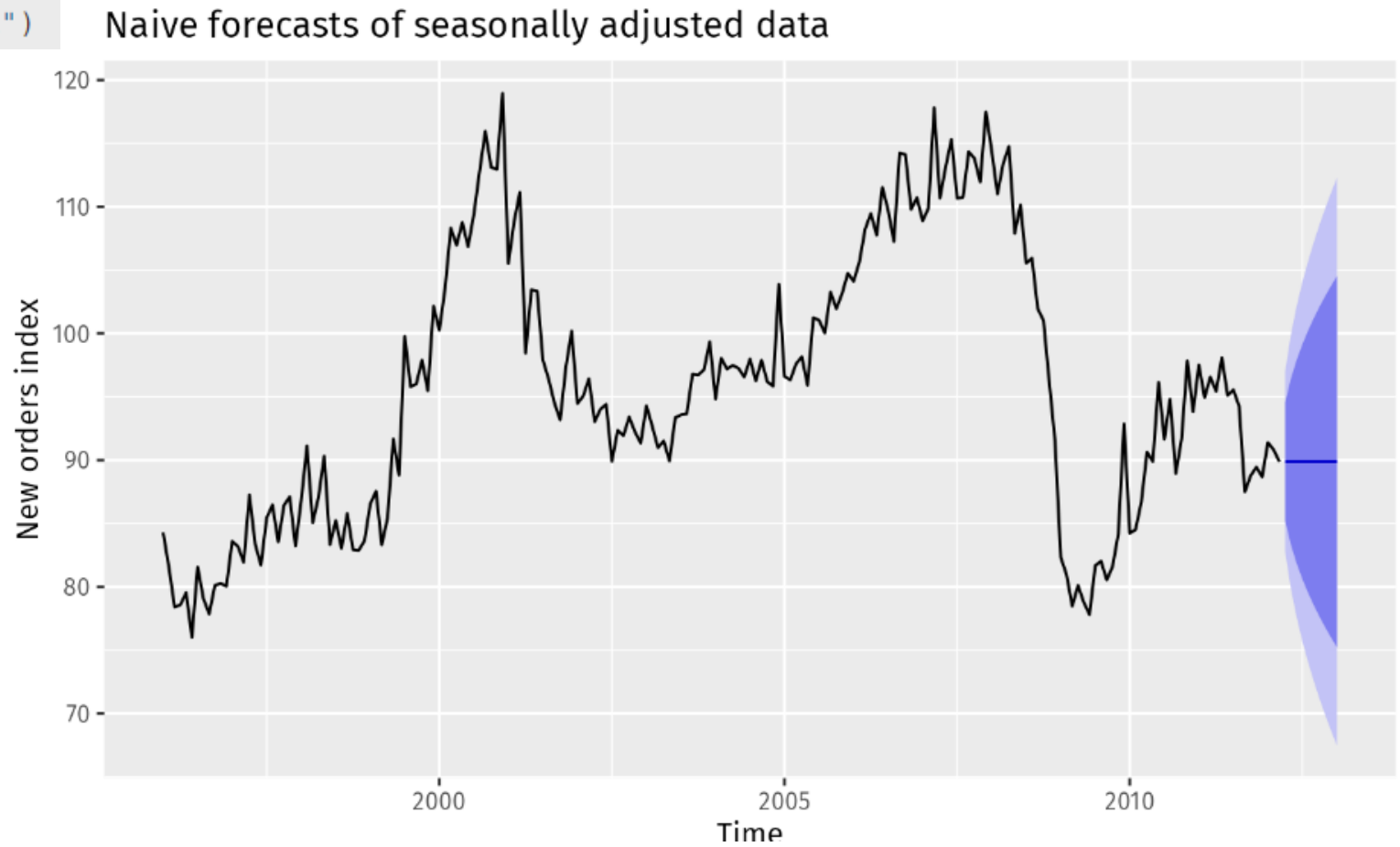
$$\boldsymbol{y_t = \widehat{S_t}\widehat{A_t}}$$

Where $\widehat{A_t} = \widehat{T_t}\widehat{R_t}$

- To **forecast** a decomposed time series, forecast **seasonal component** $\boldsymbol{\widehat{S_t}}$ and the **seasonally adjusted component** $\boldsymbol{\widehat{A_t}}$ separately .

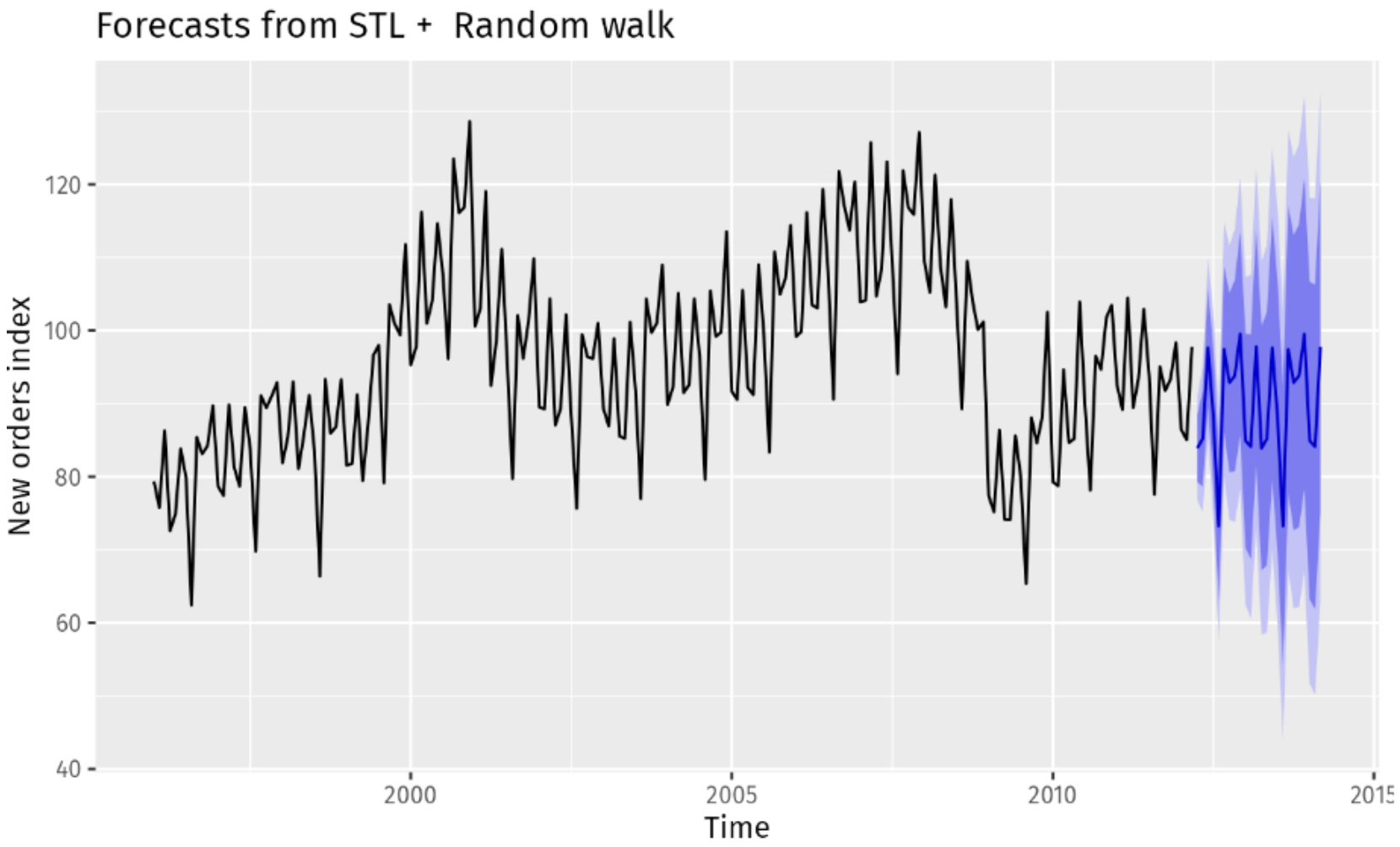
Forecasting with decomposition

```
fit <- stl(elecequip, t.window=13, s.window="periodic",
  robust=TRUE)
fit %>% seasadj() %>% naive() %>%
  autoplot() + ylab("New orders index") +
  ggtitle("Naive forecasts of seasonally adjusted data")
```



Forecasting with decomposition

```
fit %>% forecast(method="naive") %>%
  autoplot() + ylab("New orders index")
```



Review Question

1. Discuss Linear Regression Model?
2. What is coefficient of determination?
3. Show that a 3×53×5 MA is equivalent to a 7-term weighted moving average with weights of 0.067, 0.133, 0.200, 0.200, 0.200, 0.133, and 0.067.
4. Why time series decomposition is important?
5. Discuss how forecasting can be done using decomposition in a time series.

Practice Questions

2. The `plastics` data set consists of the monthly sales (in thousands) of product A for a plastics manufacturer for five years.

   a. Plot the time series of sales of product A. Can you identify seasonal fluctuations and/or a trend-cycle?
   b. Use a classical multiplicative decomposition to calculate the trend-cycle and seasonal indices.
   c. Do the results support the graphical interpretation from part a?
   d. Compute and plot the seasonally adjusted data.
   e. Change one observation to be an outlier (e.g., add 500 to one observation), and recompute the seasonally adjusted data. What is the effect of the outlier?
   f. Does it make any difference if the outlier is near the end rather than in the middle of the time series?